

KWAJALEIN ATOLL, Marshall IslandsWMO: **913660**Lat: **8.720N**Lon: **167.730E**Elev: **7**StdP: **14.69**Time Zone: **12.00 (E12)**Period: **94-19**WBAN: **40604****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	76.3	76.9	66.1	96.2	82.3	67.6	101.6	82.3	27.1	80.9	25.6	81.3	12.3	70	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.3	88.5	78.8	88.1	78.6	87.7	78.5	81.5	85.8	80.9	85.5	80.4	85.2	11.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	156.9	84.4	79.4	152.7	84.1	79.0	150.6	84.0	45.2	86.1	44.5	85.9	44.1	85.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.1	23.3	22.0	74.3	89.3	0.9	0.9	73.7	89.9	73.2	90.4	72.7	90.9	72.1	91.5		
			69.4	82.3	1.5	1.8	68.3	83.6	67.5	84.6	66.6	85.6	65.6	86.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	82.2	82.1	82.6	82.9	83.0	82.9	82.7	82.8	83.1	82.9	82.7	82.7
	DBStd	1.45	1.18	1.14	1.25	1.36	1.41	1.46	1.52	1.54	1.56	1.59	1.57	1.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11941	997	900	1011	987	1023	986	1014	1017	993	1018	982	1013
	CDD65	6466	532	480	546	537	558	536	549	552	543	553	532	548
	CDH74	74635	5819	5197	6099	6167	6514	6289	6437	6518	6513	6574	6234	6274
	CDH80	23430	1490	1306	1745	1936	2136	2061	2100	2182	2304	2235	2033	1902
Wind	WSAvg	13.4	16.8	16.9	16.2	15.8	14.4	13.3	11.2	9.2	9.0	9.6	12.5	16.7
Precipitation	PrecAvg	100.3	4.5	3.5	4.5	7.5	8.9	8.9	10.1	10.6	11.0	11.9	10.7	7.8
	PrecMax	146.5	14.4	13.8	23.4	20.3	26.9	15.7	22.3	23.5	22.0	20.0	19.2	23.3
	PrecMin	59.3	0.5	0.0	0.2	0.2	0.5	2.0	3.5	5.4	3.8	4.2	3.5	1.9
	PrecStd	20.9	3.7	3.0	4.5	5.2	5.7	3.4	3.1	4.2	3.8	4.0	4.0	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.3	87.2	88.1	88.2	88.2	88.2	88.6	88.8	89.1	89.0	88.7	87.9
		MCWB	77.0	76.8	77.4	77.2	78.0	78.7	78.9	78.9	79.2	79.3	79.3	78.5
	2%	DB	86.3	86.3	87.2	87.6	87.8	87.7	87.8	88.0	88.2	88.0	87.8	87.0
		MCWB	76.8	76.4	77.1	77.2	78.1	78.5	78.8	78.8	79.1	79.2	78.9	78.0
	5%	DB	85.7	85.9	86.3	86.7	87.0	87.0	87.3	87.6	87.6	87.3	87.0	86.3
		MCWB	76.7	76.3	77.0	77.3	78.1	78.5	78.8	78.8	79.1	79.0	78.8	77.7
	10%	DB	84.9	85.1	85.7	86.1	86.2	86.1	86.2	86.4	86.7	86.4	86.1	85.7
		MCWB	76.6	76.2	76.9	77.4	78.2	78.5	78.7	78.7	78.9	78.8	78.7	77.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	80.4	80.4	80.8	81.2	81.2	81.5	81.8	83.1	81.8	81.6	81.4
		MCDB	85.0	84.9	85.8	85.6	85.8	85.5	85.7	86.4	86.9	85.9	85.7	85.2
	2%	WB	79.6	79.1	79.2	79.7	80.4	80.5	80.6	80.8	81.0	80.8	80.7	80.3
		MCDB	84.0	83.8	84.6	84.8	85.1	85.2	85.3	85.7	85.7	85.5	85.3	84.4
	5%	WB	78.8	78.3	78.6	79.1	79.8	79.9	80.0	80.1	80.1	80.1	80.0	79.6
		MCDB	83.3	83.2	84.1	84.3	84.6	84.7	84.8	85.0	85.1	85.0	84.7	84.0
	10%	WB	78.2	77.7	78.0	78.6	79.2	79.3	79.5	79.6	79.7	79.6	79.5	79.0
		MCDB	82.8	82.8	83.6	83.9	84.1	84.3	84.4	84.5	84.8	84.6	84.3	83.5
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.3	6.6	6.5	6.5	6.6	6.9	7.2	7.3	7.1	6.7	6.0
		MCDBR	6.3	6.7	7.0	6.9	6.9	7.1	7.4	7.7	7.8	7.7	7.3	6.4
		MCWBR	3.7	3.6	3.4	3.4	3.2	3.3	3.6	3.5	3.8	3.7	3.7	3.5
	5% WB	MCDBR	6.2	6.4	6.7	6.7	6.7	6.8	7.0	7.4	7.5	7.4	7.0	6.3
		MCWBR	4.1	4.0	3.8	3.9	3.8	3.7	4.1	4.2	4.3	4.1	4.1	4.0
Clear-Sky Solar Irradiance	taub	0.439	0.443	0.445	0.456	0.437	0.428	0.415	0.394	0.394	0.399	0.437	0.457	
	taud	2.385	2.374	2.395	2.383	2.467	2.499	2.548	2.617	2.617	2.590	2.460	2.329	
	Ebn at Noon	277	279	279	272	272	271	276	285	290	289	276	269	
	Edh at Noon	39	40	40	40	36	34	33	31	31	32	36	40	
All-Sky Solar Radiation	RadAvg	1690	1849	1948	1927	1848	1809	1774	1820	1787	1662	1554	1596	
	RadStd	64	78	129	138	134	92	89	150	95	115	100	78	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.41	+0.27	N/A	+0.79	N/A	N/A	N/A	N/A	+187	+187
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page